

NGC 3198 Hidden-Sector Halo Test

Technical Audit, Final Decisions, and Rescue Framework for the Tav / Six-Domain Rotation-Curve
Draft

Prepared for Rovita Malcolm | Final review version 2.0 | June 19, 2026

One-page decision

Final recommendation: do not expand the cosmology claims until the NGC 3198 halo calculation is corrected, refit, and compared against standard halo models. The strongest contribution is to turn the current draft into a reproducible rotation-curve benchmark first. Tav-specific residual or resonance claims should be treated as Phase 2, after the ordinary mass model is numerically sound.

Review scope

This review is based on the screenshots and text supplied in the conversation, including the visible cored-isothermal halo formula, the stated parameter values $\rho_0 = 0.0160 \text{ M}_{\text{sun}} \text{ pc}^{-3}$ and $r_c = 8.58 \text{ kpc}$, and the plotted Figure 1. The underlying PDF, code, MCMC chains, and raw CSV were not directly inspected. Therefore, the conclusions below are a technical audit and rescue protocol, not a peer review of the whole Tav/Tau framework.

The practical goal is not to prove or disprove Tav cosmology. The practical goal is to identify the likely numerical failure, define the minimum correction, and give the collaborator a defensible path from speculative language to falsifiable galaxy-dynamics work.

Final decisions at a glance

Decision	Action
Immediate status	Treat the current Figure 1 as numerically suspect. The hidden-halo curve appears near zero even though the listed parameters predict a large contribution.
Core fix	Audit the conversion from $\text{M}_{\text{sun}} \text{ pc}^{-3}$ to $\text{M}_{\text{sun}} \text{ kpc}^{-3}$ and confirm $v_{\text{total}}^2 = v_{\text{baryon}}^2 + v_{\text{hidden}}^2$.
Claim discipline	State that this is an effective observational halo representation, not yet a first-principles Tav derivation.
Acceptance gate	No public success claim until the corrected plot, residuals, posterior table, code, data, and model comparisons are available.
Next milestone	Rebuild NGC 3198 as a clean benchmark; then extend to a SPARC subset.

1. Diagnostic summary

The draft makes a reasonable observational reduction: hidden sector -> collisionless virialized phase-fluid -> cored/pseudo-isothermal halo -> flat rotation curve. That is a testable map. The issue is that the displayed result appears inconsistent with the parameters shown in the screenshot.

The visible table gives approximately:

```
rho0 = 0.0160 M_sun pc^-3  
rc   = 8.58 kpc
```

For the stated halo formula, these values should produce a large hidden-halo circular speed. In the screenshot, however, the red hidden-halo curve appears nearly zero while the total and baryonic curves appear nearly identical. That pattern is exactly what one would expect if the density conversion was missed or if the hidden term was not actually wired into the plotted total model.

Most likely technical failure

- Density was used as $0.016 \text{ M}_{\text{sun}} \text{ kpc}^{-3}$ instead of $0.016 \text{ M}_{\text{sun}} \text{ pc}^{-3} = 1.60\text{e}7 \text{ M}_{\text{sun}} \text{ kpc}^{-3}$.
- The hidden-halo velocity was computed correctly but not added to v_{total}^2 .
- The red curve layer is plotting an unconverted density, normalized diagnostic, or placeholder instead of $v_{\text{h}}(r)$.
- The MCMC table and plotted curve may come from different code paths or stale outputs.

Confidence rating

Finding	Confidence	Reason
The red hidden curve should not be near zero for the listed rho0 and rc.	High	Direct evaluation of the displayed formula and parameter values.
A pc^3/kpc^3 density conversion bug is plausible.	High	Missing the $1\text{e}9$ conversion reduces v_{h} by $\sqrt{1\text{e}9}$, making the curve visually vanish.
The listed rho0 may overproduce the curve after correction.	Medium	The asymptotic speed from the table is about 252 km/s before adding baryons.
The Tav framework is validated by the fit.	Not established	A cored halo fit is not unique to Tav and standard halo comparisons are required.

2. Correct model and unit audit

Use the pseudo-isothermal / cored-isothermal density profile and circular-speed contribution exactly as follows:

$$\rho_{\text{h}}(r) = \rho_0 / [1 + (r/r_c)^2]$$

$$v_{\text{h}}^2(r) = 4\pi G \rho_0 r_c^2 * [1 - (r_c/r) * \text{atan}(r/r_c)]$$

$$v_{\text{model}}^2(r) = v_{\text{gas}}^2(r) + \text{Upsilon}_{\text{disk}} v_{\text{disk}}^2(r) + \text{Upsilon}_{\text{bulge}} v_{\text{bulge}}^2(r) + v_{\text{h}}^2(r)$$

Use $G = 4.30091\text{e-}6 \text{ kpc (km/s)}^2 \text{ M}_{\text{sun}}^{-1}$. If rho0 is quoted in $\text{M}_{\text{sun}} \text{ pc}^{-3}$, it must be converted before insertion into a kpc-based formula:

$$\rho_{0 \text{ kpc}} = \rho_{0 \text{ pc}} * 1\text{e}9$$

With the screenshot values, $v_{\text{infinity}} = \sqrt{4\pi G \rho_{0 \text{ kpc}} r_c^2} = 252.3 \text{ km/s}$. That is not a small number. A red halo curve near zero is incompatible with the table and formula unless the plot is not the true halo velocity.

Numerical audit table

r (kpc)	v_h with correct conversion (km/s)	v_h if conversion missed (km/s)
1	16.9	0.00053
2	33.4	0.00106
5	77.6	0.00245
8.58	116.9	0.00370
10	128.8	0.00407
15	159.3	0.00504
20	178.4	0.00564
30	200.3	0.00634
48	218.6	0.00691

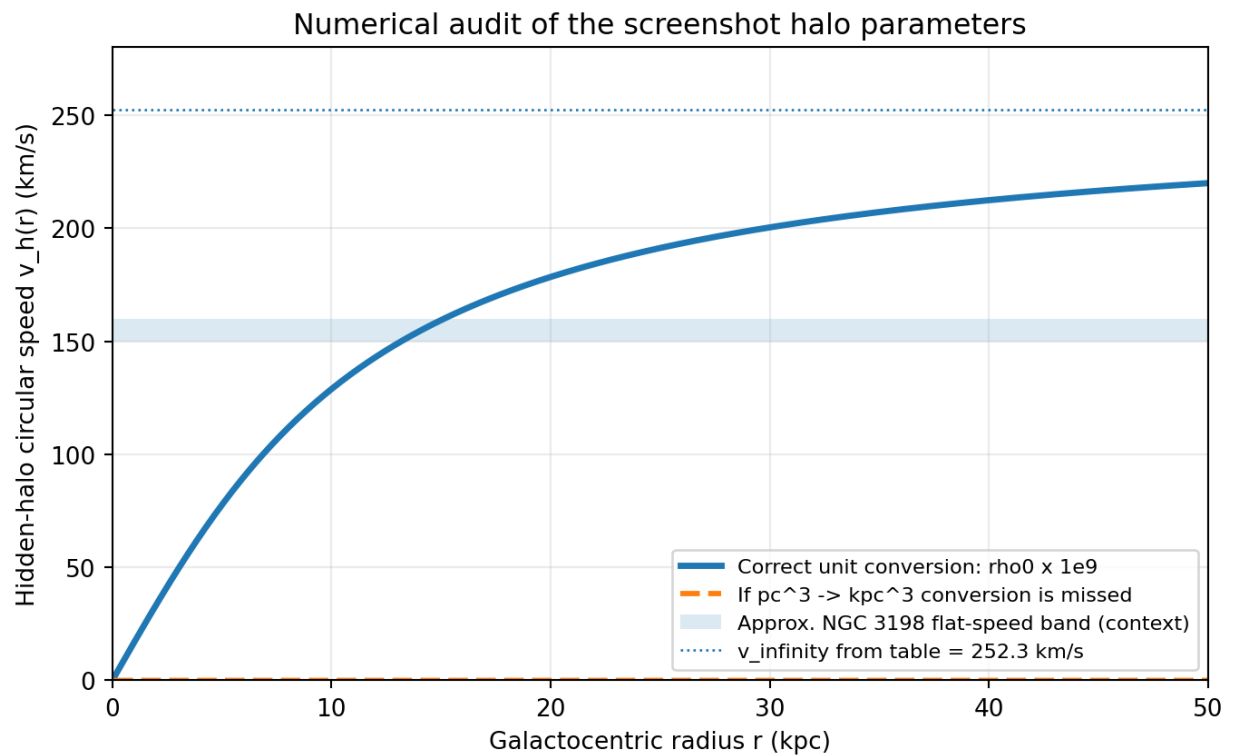


Figure 1. The listed screenshot parameters produce a large halo velocity when units are correct. If the pc^3 to kpc^3 density conversion is missed, the halo curve becomes effectively zero - matching the visual failure suggested by the screenshot.

3. Second-order issue after the bug is fixed

Correcting the hidden-halo curve may expose a different problem: the posterior table itself may be invalid or stale. If NGC 3198's outer rotation speed is roughly 150-160 km/s and the baryonic contribution is nonzero, the screenshot value $\rho_0 = 0.0160 \text{ M}_{\text{sun}} \text{ pc}^{-3}$ with $r_c = 8.58 \text{ kpc}$ is likely too large because it gives v_h around 200 km/s at 30 kpc and an asymptotic speed near 252 km/s before baryons are added in quadrature.

At the same $r_c = 8.58 \text{ kpc}$, a rough density giving $v_{\text{infinity}} = 155 \text{ km/s}$ is $\rho_0 \approx 0.0060 \text{ M}_{\text{sun}} \text{ pc}^{-3}$. This is a back-of-envelope target only, not a fit. The correct action is to refit ρ_0 , r_c , and mass-to-light ratio after the calculation is repaired.

Critical distinction

Question	Answer
Does the audit solve the full galaxy fit?	No. It identifies a likely numerical bug and the required refit protocol.
Does a corrected pseudo-isothermal halo prove Tav?	No. It proves only that a standard cored halo profile can be fit or tested in this context.
What would make the model Tav-specific?	A derived relation for ρ_0 , r_c , normalization, or residual structure from Tav parameters, or a statistically significant residual signature not explained by standard alternatives.
What is the immediate win?	One corrected, reproducible, hard-to-dismiss NGC 3198 benchmark with honest comparisons.

4. Final decisions

These are the final decisions for how to proceed.

Decision	Required action
D1 - Freeze the scope	The current paper should claim only an effective hidden-sector halo test for NGC 3198, not a solved dark-matter theory.
D2 - Treat Figure 1 as failed until regenerated	The plotted hidden curve appears inconsistent with the listed ρ_0 and r_c . Regeneration from clean code is mandatory.
D3 - Audit units first	Every kpc-based calculation must use $\rho_{0\text{ kpc}} = \rho_{0\text{ pc}} * 1e9$. Add a unit test that fails if the conversion is removed.
D4 - Audit quadrature	The model must add squared velocities, not velocities, densities, or accelerations: $v_{\text{total}}^2 = \text{sum of component } v_i^2$.
D5 - Refit after repair	Do not reuse the posterior table unless it is generated from the corrected code and supported by residuals.
D6 - Compare standard halos	Pure baryons, NFW, Burkert, and ordinary pseudo-isothermal halos are mandatory baselines.
D7 - Separate effective mapping from first-principles derivation	It is acceptable to use a cored halo as an observational representation. It is not acceptable to say T_{av} derives ρ_0 and r_c unless the derivation is shown.
D8 - Move T_{av} harmonics to Phase 2	A 1/7 or log-periodic residual test should be pre-registered and run only after the smooth halo model is correct.
D9 - Require reproducibility	Public CSV, code, environment, residual plots, and figure-regeneration commands are acceptance requirements.
D10 - Expand only after one clean benchmark	After NGC 3198 is fixed, run the same pipeline on a small SPARC subset before making broader claims.

5. Rescue framework: Hidden-Sector Rotation-Curve Pipeline v1.0

This pipeline converts the draft into a controlled, auditable galaxy-dynamics project.

Stage	Purpose	Minimum output
0. Scope lock	Prevent theory sprawl	One claim, one galaxy, one profile, one repository.
1. Data ingestion	Use public rotation-curve data	CSV with r , V_{obs} , eV_{obs} , V_{gas} , V_{disk} , V_{bulge} where available.
2. Unit audit	Prevent hidden-term collapse	Automated tests for ρ_0 conversion and velocity scale.
3. Baryonic baseline	Show the missing-mass problem honestly	Baryons-only plot, residuals, reduced chi-square.
4. Effective halo fit	Test the hidden-sector representation	Posterior table for ρ_0 , r_c , stellar M/L, nuisance parameters.
5. Model comparison	Stop false uniqueness	AIC/BIC or Bayes factors vs baryons-only, NFW, Burkert, pISO.
6. Residual analysis	Find what remains unexplained	$v_{\text{obs}} - v_{\text{model}}$ panel with uncertainties and systematics notes.
7. T_{av} Phase 2	Test unique resonance claim	Pre-registered log-periodic residual model and null comparison.
8. SPARC extension	Avoid one-galaxy cherry-picking	Same pipeline on a small subset, then all SPARC if justified.

Minimum acceptance criteria

- The corrected hidden-halo curve is visible and numerically matches the formula.
- The total model is generated from the same arrays used for residuals and fit statistics.
- The residual panel includes observational uncertainties and a zero reference line.
- The posterior table is reproducible from public code and data.
- The paper reports whether standard halos fit equally well or better.
- The final claims avoid language such as 'proved', 'confirmed', or 'zero free parameters' unless those statements are demonstrated.

6. Implementation protocol

Repository structure

```
data/ngc3198_rotation_curve.csv
src/halo_models.py
src/fit_ngc3198.py
src/model_comparison.py
tests/test_units.py
figures/ngc3198_corrected_fit.png
figures/ngc3198_residuals.png
figures/ngc3198_model_comparison.png
README.md
environment.yml or requirements.txt
```

Unit-test pseudocode

```
import numpy as np

G = 4.30091e-6 # kpc (km/s)^2 / M_sun

def vh_piso(r_kpc, rho0_pc3, rc_kpc):
    rho0_kpc3 = rho0_pc3 * 1e9
    term = 1.0 - (rc_kpc/r_kpc) * np.arctan(r_kpc/rc_kpc)
    return np.sqrt(4*np.pi*G*rho0_kpc3*rc_kpc**2*term)

def test_screenshot_halo_not_zero():
    assert 120.0 < vh_piso(10.0, 0.0160, 8.58) < 140.0
    assert 190.0 < vh_piso(30.0, 0.0160, 8.58) < 210.0

def test_missing_conversion_is_detected():
    # If rho0 is treated as M_sun/kpc^3, v_h is effectively zero.
    rho_wrong = 0.0160
    rc = 8.58
    r = 10.0
    term = 1.0 - (rc/r) * np.arctan(r/rc)
    v_wrong = np.sqrt(4*np.pi*G*rho_wrong*rc**2*term)
    assert v_wrong < 0.01
```

Model comparison rules

Model	Why it must be included	Minimum output
Baryons only	Shows the missing-mass problem in the chosen galaxy.	chi-square, residuals.
NFW	Standard Lambda-CDM inspired comparison.	best-fit parameters, fit quality, AIC/BIC.
Burkert	Common cored-halo comparison.	best-fit parameters, fit quality, AIC/BIC.
Ordinary pISO	Same empirical shape without Tav interpretation.	fit quality and posterior.
Tav-hidden pISO	Same profile interpreted as hidden phase-fluid.	Not unique unless parameters are derived.
Tav residual harmonic	Potentially unique Phase 2 claim.	Bayes factor vs smooth residual null.

7. Tav-specific Phase 2 test

Only after the smooth model is correct should the team test Tav-specific structure. A possible residual model is:

$$\text{delta_v}(r) = A * \sin(2*\pi*\ln(r/r0)/\ln(7) + \text{phi})$$

This model should be fitted to residuals after subtracting baryons plus the smooth cored-isothermal halo. It must be compared to a null residual model and to generic sinusoidal/log-periodic alternatives. The test should be pre-registered: fix the frequency, radial range, allowed amplitude prior, data-quality cuts, and significance threshold before searching for a detection.

A positive residual detection is interesting only if it survives look-elsewhere correction, distance/inclination uncertainty, beam-smearing issues, gas asymmetries, spiral arms, bars, and ordinary non-circular motions. A null result would not kill the entire framework; it would place an upper limit on local Tav coupling strength.

Acceptable theoretical bridge for the current paper

six-domain hidden sector
-> collisionless hidden phase-fluid
-> virialized quasi-isothermal equilibrium
-> finite-core pseudo-isothermal density profile
-> circular-speed contribution added in quadrature to baryons
-> rotation-curve fit and residual test

This bridge is acceptable as a phenomenological mapping. It is not yet a first-principles derivation unless ρ_0 , r_c , and the normalization are derived from a defined T_{av} action or microphysical hidden-sector model.

Protective sentence to include in the paper

At this stage, the cored-isothermal profile is used as an effective observational representation of the hidden-sector phase-fluid. Deriving the halo parameters directly from the T_{av} action is a separate theoretical task.

8. Rewrite guidance for the paper

Recommended title

Effective Hidden-Sector Cored-Halo Benchmark for NGC 3198: A Reproducible Rotation-Curve Test

Recommended abstract language

We test whether the Tav/Six-Domain hidden sector can be represented observationally as an effective collisionless, virialized, cored-isothermal phase-fluid halo. Using the NGC 3198 rotation curve as a benchmark, we fit a baryonic mass model plus a pseudo-isothermal halo contribution and compare the result against baryons-only and standard halo alternatives. This paper does not claim a first-principles derivation of halo parameters from Tav geometry; it defines a reproducible observational reduction and the conditions under which Tav-specific residual tests can be attempted.

Claims to remove or soften

Avoid	Replace with
Tav dark matter is confirmed.	The effective hidden-sector halo representation can be tested against NGC 3198.
Zero free parameters.	Parameters are fitted or assigned priors unless explicitly derived.
The six-domain structure proves the halo.	The six-domain structure motivates an effective phase-fluid representation.
The model reproduces the curve.	The model reproduces the curve after unit audit, refit, and residual validation.
Standard dark matter fails.	We compare the effective model to NFW, Burkert, and pseudo-isothermal baselines.

Draft message to send the collaborator

I think the NGC 3198 draft is on the right observational target, but Figure 1 appears internally inconsistent. The table gives $\rho_0 = 0.0160 \text{ M}_{\text{sun}} \text{ pc}^{-3}$ and $r_c = 8.58 \text{ kpc}$. If those values are inserted into the cored-isothermal velocity formula with the required conversion from $\text{M}_{\text{sun}} \text{ pc}^{-3}$ to $\text{M}_{\text{sun}} \text{ kpc}^{-3}$, the hidden halo contribution should be large - roughly 129 km/s at 10 kpc and 200 km/s at 30 kpc. In the screenshot, the red hidden-halo curve appears nearly zero. Please audit whether the density conversion factor of $1e9$ was missed, and confirm that the model uses $v_{\text{total}}^2 = v_{\text{baryon}}^2 + v_{\text{hidden}}^2$. After that, the parameters should be refit, because the current ρ_0 may overproduce the curve once the halo term is correctly included. The safest claim for this draft is not that Tav dark matter is proven, but that the six-domain hidden sector can be tested observationally as an effective cored-isothermal phase-fluid halo. The paper becomes much stronger if it includes a corrected fit, residual panel, posterior table, unit test, public code/data, and comparison with baryons-only, NFW, Burkert, and ordinary pseudo-isothermal halos.

9. Risk register and triage outcomes

After correction	Meaning	Decision
Hidden curve appears and total fits	The bug was probably plotting/units; the model may be salvageable.	Proceed to residual plot and model comparison.
Hidden curve appears but overproduces speeds	The table/posterior is invalid or priors are wrong.	Refit ρ_0 , r_c , and stellar M/L.
Hidden curve remains zero	Halo term is still not wired into plot/calculation.	Debug v_h function and plotting arrays.
Fit requires extreme M/L or halo parameters	Model is tuned or physically strained.	Report honestly; compare to standard halos.
Standard halos fit equally well	Tav is not empirically distinguished yet.	Move uniqueness to Phase 2 residual or derivation.
Residual harmonic appears	Potential Tav-specific signal, but high false-positive risk.	Run pre-registered null tests and systematics checks.

Final implementation checklist

Item	Pass condition
Unit conversion	$\rho_{0,pc}$ is multiplied by $1e9$ before entering the kpc velocity formula.
Velocity formula	v_h^2 uses the pseudo-isothermal expression exactly.
Quadrature	v_{total}^2 is the sum of squared velocity components.
Plot	Observed data, gas, disk, hidden halo, total, and uncertainty band are visually distinct.
Residuals	Residual panel includes zero line and observational uncertainties.
Posteriors	ρ_0 , r_c , stellar M/L, distance/inclination if varied, and nuisance parameters have intervals.
Comparisons	Baryons-only, NFW, Burkert, and ordinary pISO benchmarks are included.
Claims	Paper states this is an effective mapping unless T_{av} parameters are derived.
Reproducibility	A reader can regenerate every figure from public code and data.

10. Source anchors, limitations, and bottom line

Public source anchors used to ground the non-Tav galaxy-dynamics context:

Source	Why it matters
SPARC official database; Lelli, McGaugh & Schombert 2016	SPARC provides a public benchmark of 175 nearby late-type galaxies with Spitzer 3.6 micron photometry and high-quality HI/Halpha rotation curves.
van Albada et al. 1985, ApJ	Classic NGC 3198 disk-plus-halo rotation-curve analysis; dark matter inside the last measured point was found to exceed visible matter by a large factor.
Begeman 1989, A&A;	NGC 3198 HI rotation curve extends far into the outer disk and is a classic flat-curve case.
Karukes, Salucci & Gentile 2015	Extended HI kinematics to about 48 kpc and comparisons of halo models for NGC 3198.
Li, Lelli, McGaugh & Schombert 2020	Shows why comparing multiple halo models across SPARC is standard and necessary.
Salucci/NED dark-halo profile review	Pseudo-isothermal halos are standard finite-core profiles that tend toward flat circular speeds at large radius.

Limitations

- The underlying draft PDF, code, MCMC chains, and raw CSV were not directly inspected; screenshots are the basis of the numerical diagnosis.
- The audit is exact for the formula and parameters visible in the screenshots; it is not a full refit to SPARC or HALOGAS data.
- The Tav/Tau framework remains speculative unless distinctive parameters or residual signatures are derived and tested against alternatives.
- This memo should be used as a technical rescue plan and collaborator communication guide, not as a claim of proof.

Final bottom line

The best help is to make one plot that cannot be dismissed. Fix the unit conversion, wire the hidden velocity term correctly, refit NGC 3198, add residuals, compare to standard halos, publish code/data, and postpone broad Tav claims until the empirical benchmark survives. That is the path from scattered theory into a serious falsifiable program.